

AICTE | IBM SkillsBuild

SUPERMARKET SALES ANALYSIS WITH AI

Data Analytics Internship Project Report

Submitted by

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CERTIFICATE / DECLARATION

This project report titled “Supermarket Sales Analysis with AI” documents the analytical implementation prepared for academic and internship presentation purposes. The project demonstrates a complete workflow from transaction data preparation to exploratory analysis, machine-learning based anomaly detection, dashboard visualization, and business interpretation.

Student Declaration

I, Shraddha Dhakad, declare that this report is prepared as an academic project based on the supermarket-sales analytics implementation supplied with this submission. The numerical results in this report are calculated from the included demonstration CSV dataset. The attached internship resource identifies the Masterclass 4 workbook as “Supermarket Sales Analysis DA project”; the official workbook data itself was not contained in the supplied PDF. Therefore, this report clearly distinguishes the demonstration dataset from any official internship dataset that may be provided separately.

Signature: _____

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the organizations and mentors associated with the AICTE | IBM SkillsBuild Data Analytics with AI Internship Program 2026 for providing a structured learning environment covering data fundamentals, data cleaning and preparation, AI for data analytics, and business-oriented solutions.

I also acknowledge the value of the provided session resources and project guidance in shaping this implementation. The supplied resource document records BharatCares as the company, the six-week online internship period, and Mr. Kartik Hooda as trainer. These details are reproduced only as project context; the analysis itself is based on the demonstration dataset bundled with this project.

Finally, I thank my college, Bansal College of Engineering, Mandideep, and the faculty community for encouraging practical application of Python, data analytics, and machine learning concepts.

Project Focus

- Data preparation and validation
- Exploratory data analysis and KPI reporting
- Machine-learning based anomaly detection
- Customer-level summary and payment analysis
- Streamlit dashboard and business insight generation

ABSTRACT

Retail transactions contain useful information about revenue, products, branches, customer groups, payment preferences, and unusual purchase patterns. The Supermarket Sales Analysis with AI project develops an end-to-end analytical workflow that converts transaction data into a compact decision-support dashboard. The implementation uses Python, Pandas, NumPy, Plotly, Scikit-learn, and Streamlit.

The workflow starts with data validation and cleaning. Missing numerical values are repaired using median imputation, categorical blanks are mapped to “Unknown”, and derived fields are created for revenue before tax, month, day name, and profit margin. Exploratory analysis then calculates key performance indicators, compares branches and product lines, studies customer and payment behavior, and visualizes time trends.

The AI component uses Isolation Forest to identify statistically unusual transactions. A customer-summary layer also provides a segmentation scaffold; because the demonstration data contains only two customer types, the implemented code intentionally uses stable labels rather than forcing three K-Means clusters. The final dashboard presents the findings in an interactive Streamlit interface and exposes cleaned data for download.

Keywords: Data Analytics, Retail Analytics, Supermarket Sales, Python, Pandas, Machine Learning, Isolation Forest, K-Means, Streamlit, Business Intelligence

TABLE OF CONTENTS

1. Internship Context and Project Alignment.....	6
2. Introduction.....	7
3. Problem Statement and Motivation.....	8
4. Objectives and Scope.....	9
5. Dataset Description.....	10
6. Data Dictionary.....	11
7. Methodology.....	12
8. Data Preprocessing.....	13
9. Exploratory Data Analysis and KPIs.....	14
10. Branch Performance Analysis.....	15
11. Product Line Analysis.....	16
12. Customer and Payment Analysis.....	17-18
13. AI-Based Anomaly Detection.....	19
14. Customer Segmentation.....	20
15. System Architecture and Dashboard.....	21
16. Results and Business Insights.....	22
17. Limitations.....	23
18. Future Scope and Conclusion.....	24
19. References and Appendix.....	25

1. INTERNSHIP CONTEXT AND PROJECT ALIGNMENT

The supplied internship resource is titled “AICTE | IBM SkillsBuild Data Analytics with AI Internship Program 2026” and describes a six-week online program from 17 August to 30 September 2026. The stated focus is “Data Analytics with AI: Foundation to Implementation”.

Program item	Source-derived detail
Company	BharatCares
Duration	6 weeks; 17 Aug - 30 Sep 2026
Mode	Online
Focus	Data Analytics with AI: Foundation to Implementation
Trainer	Mr. Kartik Hooda
Project resource	Supermarket Sales Analysis DA project

The resource schedule includes sessions on Data Fundamentals, Data Cleaning & Preparation, AI for Data Analytics, Business Solutions with IBM BOB, a Project Discussion, and Career Connect. This implementation follows the same broad progression: understand the data, prepare it, analyze it, apply AI, then communicate business-oriented outputs.

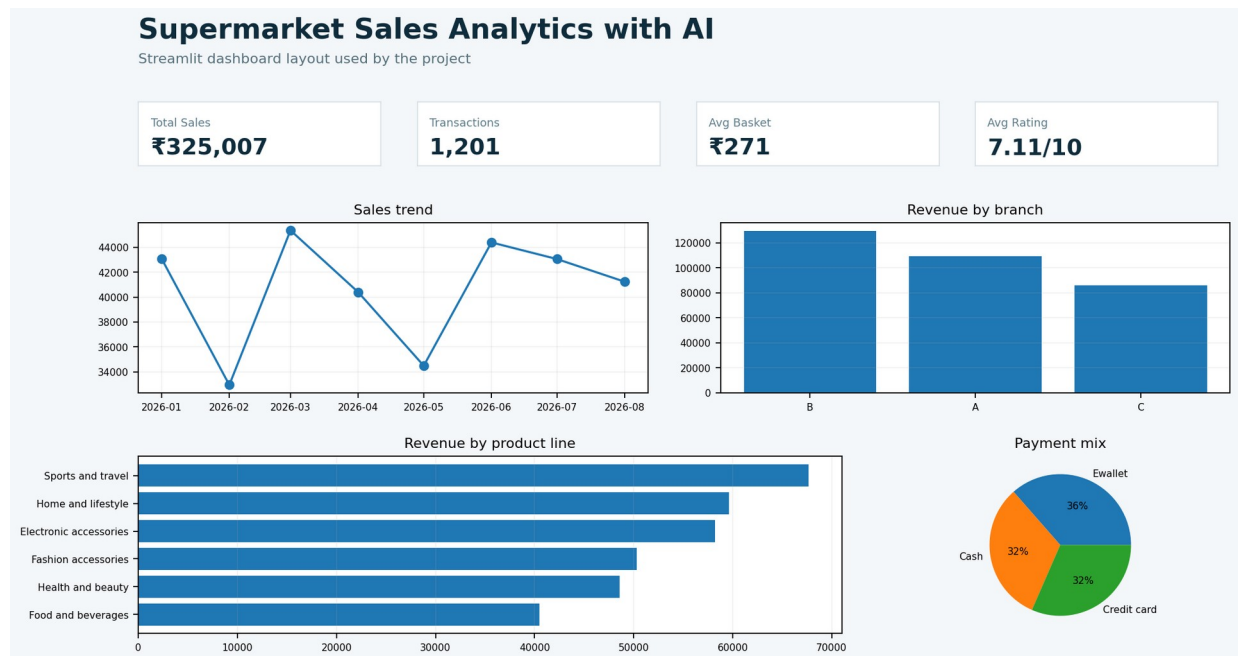


Figure 1. High-level view of the Streamlit dashboard implemented in the project.

2. INTRODUCTION

Modern retail operations produce transaction records continuously. A useful analytics system should reduce the gap between these records and decisions. Instead of reading hundreds or thousands of invoices individually, a manager can use aggregated KPIs, trend charts, branch comparisons, product rankings, payment distributions, and anomaly alerts.

This project demonstrates that workflow on a supermarket-sales problem. The application is intentionally beginner-friendly and modular. Data loading and cleaning are isolated in `preprocessing.py`, machine-learning logic is isolated in `ai_models.py`, and the Streamlit user interface is implemented in `app.py`. This separation makes the project easier to understand, modify, and extend.

The analytical emphasis is descriptive rather than causal. The dashboard shows what happened in the data and identifies statistically unusual records; it does not prove why sales changed or why a transaction is unusual. This distinction is important when analytics are used in real operational environments.

The project also follows a practical principle: business communication matters as much as model selection. A complex model is less useful in an internship setting if its output cannot be explained clearly. For that reason, AI results are translated into plain-language insights and simple tables.

3. PROBLEM STATEMENT AND MOTIVATION

Problem Statement

Raw supermarket transaction data is difficult to interpret manually. The project therefore asks: how can transaction-level records be cleaned, summarized, visualized, and augmented with machine-learning methods so that a store manager can quickly understand sales performance and investigate unusual transactions?

Motivation

- Large transaction tables contain useful patterns but hide them inside row-level detail.
- Branch and product differences matter for resource allocation, promotions, and inventory review.
- Payment and customer-type distributions help describe shopper behavior.
- Statistical anomaly detection can prioritize transactions for human review.
- An interactive dashboard makes the analysis easier to communicate during internship and viva presentations.

The motivation is therefore not to replace human decisions with AI. It is to make the relevant evidence easier to find and discuss.

4. OBJECTIVES AND SCOPE

Project Objectives

- Validate the structure of a supermarket transaction dataset.
- Repair selected missing values and create analysis-ready features.
- Calculate sales, transaction count, average basket, and rating KPIs.
- Compare revenue across branches and product lines.
- Summarize customer type and payment behavior.
- Apply Isolation Forest for statistical anomaly detection.
- Provide a customer-summary segmentation layer without overfitting a small number of customer groups.
- Present results through a responsive Streamlit dashboard and export cleaned data.

Scope

The current project is a supervised academic/internship demonstration. It uses a local CSV, descriptive analytics, and lightweight machine-learning components. It does not include a transactional database, authentication, automated retraining, demand forecasting, causal analysis, or live business data integration.

Out of scope

- Fraud determination
- Financial forecasting guarantees
- Customer identity resolution
- Real-time POS integration
- Production-grade security controls

5. DATASET DESCRIPTION

The package contains a generated demonstration CSV named `supermarket_sales.csv` with 1,201 transaction records and 16 original columns. The data spans 240 dates in 2026 and includes three branches/cities, two customer types, two genders, six product lines, three payment methods, and ratings on a 0-10 scale.

Important source limitation: the attached internship resource names the Masterclass 4 workbook “Supermarket Sales Analysis DA project”, but the supplied four-page PDF does not include the workbook data itself. Therefore, the numbers in this report describe the demonstration dataset bundled with the code package. They should not be presented as official internship-business results unless the official workbook is substituted and the analysis rerun.

Dimension	Value
Rows	1,201
Original columns	16
Branches	A, B, C
Cities	Bhopal, Indore, Jabalpur
Customer types	Member, Normal
Product lines	6
Payment methods	Cash, Credit card, Ewallet
Date range	Jan-Aug 2026
Rating range	4.0-10.0

Primary analytical variables include Total, Quantity, Unit_Price, COGS, Gross_Income, Gross_Margin_pct, Payment, Rating, Branch, Product_Line, and Customer_Type.

6. DATA DICTIONARY

Field	Type	Meaning
Invoice_ID	Text	Unique transaction identifier
Date	Date	Transaction date
Branch	Category	Store branch code
City	Category	Branch city
Customer_Type	Category	Member or Normal
Gender	Category	Customer gender label
Product_Line	Category	Product category
Unit_Price	Numeric	Price per unit
Quantity	Numeric	Units purchased
Tax_5pct	Numeric	5% tax amount
Total	Numeric	Transaction total
COGS	Numeric	Cost of goods sold
Gross_Margin_pct	Numeric	Gross margin percentage
Gross_Income	Numeric	Gross income
Payment	Category	Payment method
Rating	Numeric	Customer rating

Derived fields created by preprocessing.py

Derived field	Purpose
Revenue_Before_Tax	$\text{Unit_Price} \times \text{Quantity}$
Month	Monthly aggregation key derived from Date
Day_Name	Day-of-week label derived from Date
Profit_Margin_pct	$\text{Gross_Income} / \text{Total} \times 100$

7. METHODOLOGY

The project follows a repeatable analytics pipeline.

Step	Stage	Implementation
1	Data loading	Read the CSV and validate that required columns are present.
2	Type conversion	Convert Date and analytical measures to appropriate numeric/date types.
3	Quality repair	Remove duplicate invoice identifiers and impute selected missing numeric values.
4	Feature creation	Create revenue-before-tax, month, day-name, and profit-margin fields.
5	EDA	Compute KPIs and grouped summaries for branch, product, customer, payment, and time dimensions.
6	AI analysis	Run Isolation Forest on standardized transaction features and provide a customer-segmentation scaffold.
7	Communication	Render results as a Streamlit dashboard and plain-language insights.
8	Export	Allow the filtered, cleaned dataset to be downloaded as CSV.

Design principle: each stage produces an output that can be checked before moving to the next stage. This improves transparency and makes the project suitable for academic demonstration.

8. DATA PREPROCESSING

The raw dataset contains 1,201 rows. Two missing values were detected in the fields Quantity and Rating. The implemented preprocessing routine fills these fields using their respective medians, rounds Quantity to an integer, clips it to at least one unit, and constrains Rating to the 0-10 range.

Quality check	Before	Action	After
Rows	1,201	Retain after validation	1,201
Duplicate invoice IDs	0	No duplicates removed	0
Missing values	2	Median imputation	0
Quantity type	Numeric/float	Round + cast to int	Integer
Rating scale	Numeric	Clip to 0-10	Valid range

The code also checks for all 16 required columns before analysis. Categorical missing values, if present, are converted to “Unknown”. This approach is practical for a teaching project because it is easy to explain and audit.

Data-cleaning caution: median imputation preserves row count but changes the exact value of an incomplete record. In production, the replacement rule should be documented alongside the data lineage and, where possible, the original raw record should remain preserved for audit.

9. EXPLORATORY DATA ANALYSIS AND KPIS

After cleaning, the dataset contains 1,201 usable transactions. Total sales are ₹325,006.87, average transaction value is ₹270.61, and the mean customer rating is 7.11/10. The average daily sales value in the latest seven dates is 7.4% lower than the preceding seven dates represented in the generated dataset.

KPI	Value	Interpretation
Total sales	₹325,006.87	Sum of transaction totals
Transactions	1,201	Number of transaction records
Average basket	₹270.61	Mean transaction value
Average rating	7.11 / 10	Mean customer rating
Flagged anomalies	36 (3.0%)	Statistical review candidates

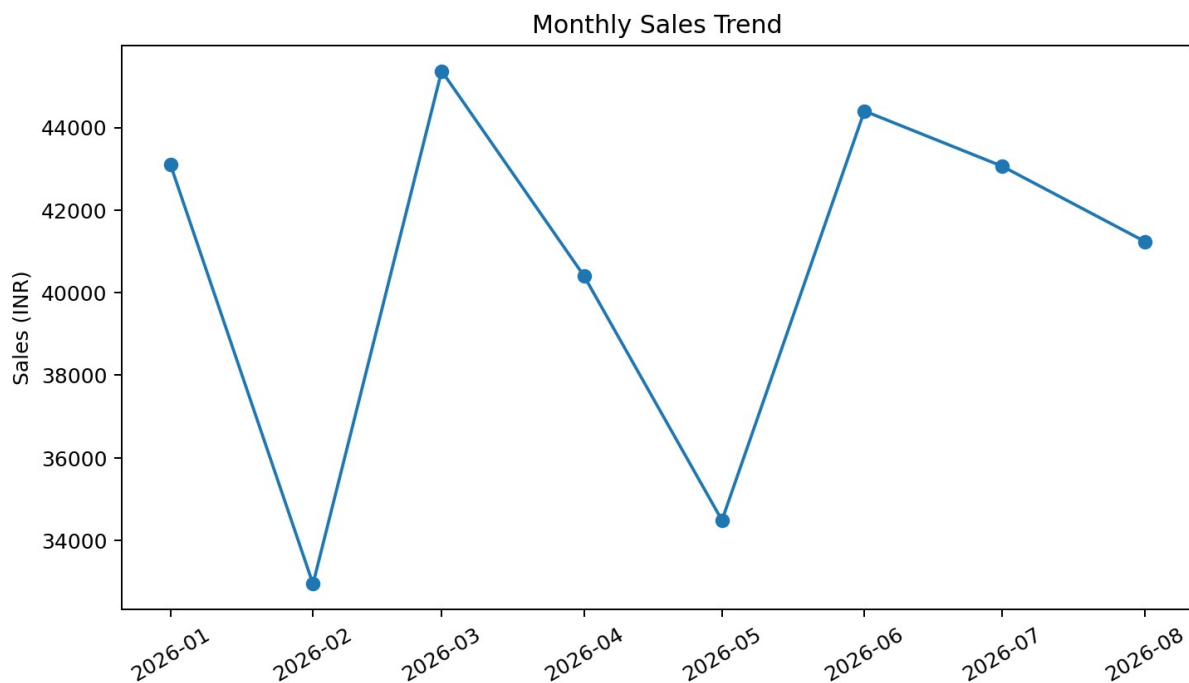


Figure 2. Monthly sales trend for the demonstration dataset.

EDA is descriptive: these KPIs summarize observed records. They should not be interpreted as forecasts or causal explanations.

10. BRANCH PERFORMANCE ANALYSIS

Branch analysis compares transaction count, total sales, average basket, and gross income across the three branch-city combinations.

Branch	City	Txns	Sales	Avg basket	Gross income
B	Indore	474	₹129,669.80	₹273.56	₹44,151.96
A	Bhopal	414	₹109,323.44	₹264.07	₹36,404.80
C	Jabalpur	313	₹86,013.63	₹274.80	₹28,795.80

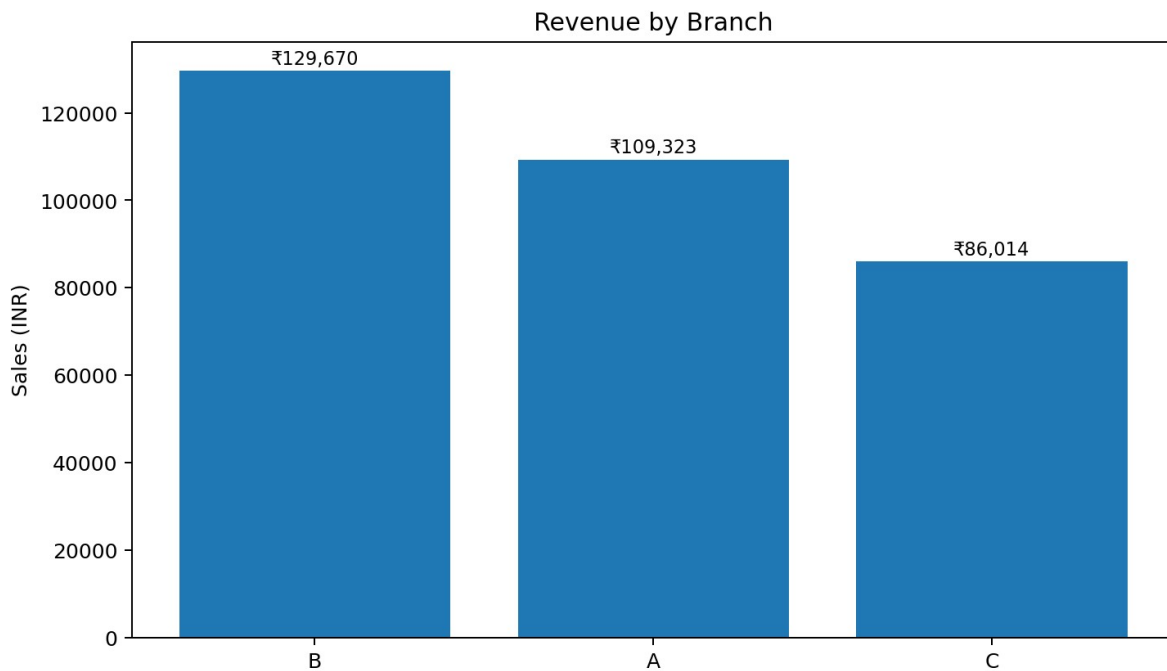


Figure 3. Revenue comparison by branch.

Observation: Branch B (Indore) records the highest total sales in this dataset at ₹129,669.80. Branch C (Jabalpur) has the lowest sales total among the three branches. These are descriptive results from the demonstration data, not recommendations about how a real retailer should allocate investment.

11. PRODUCT LINE ANALYSIS

Product-line analysis identifies categories with higher revenue and quantity contribution. The six categories are compared using transaction count, sales, quantity, and gross income.

Product line	Txns	Qty	Sales	Gross income
Sports and travel	211	1023	₹67,657.37	₹22,947.28
Home and lifestyle	191	958	₹59,659.06	₹19,578.04
Electronic accessories	205	1006	₹58,233.30	₹19,423.94
Fashion accessories	186	948	₹50,335.71	₹17,343.12
Health and beauty	195	951	₹48,610.95	₹16,614.17
Food and beverages	213	989	₹40,510.48	₹13,446.01

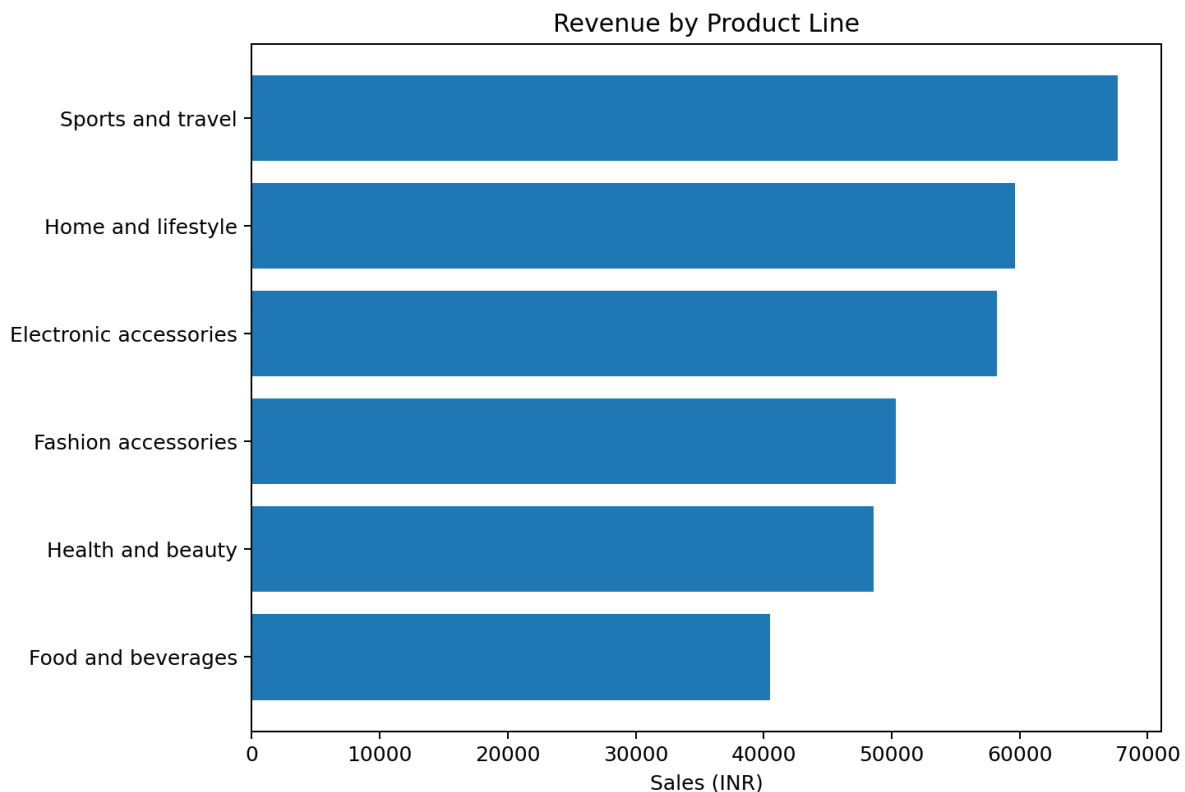


Figure 4. Revenue by product line.

Observation: Sports and travel is the highest-revenue product line at ₹67,657.37, while Food and beverages is the lowest at ₹40,510.48 in the demonstration dataset. Category revenue alone does not explain profitability, inventory turnover, or customer satisfaction; those questions require additional data.

12. CUSTOMER AND PAYMENT ANALYSIS

Customer-type summaries show that Normal customers account for 623 transactions and ₹171,722.32 in sales, while Member customers account for 578 transactions and ₹153,284.55 in sales. Member customers have a slightly higher mean rating (7.12) than Normal customers (7.10), while Normal customers show a higher mean basket value (₹275.64 versus ₹265.20).

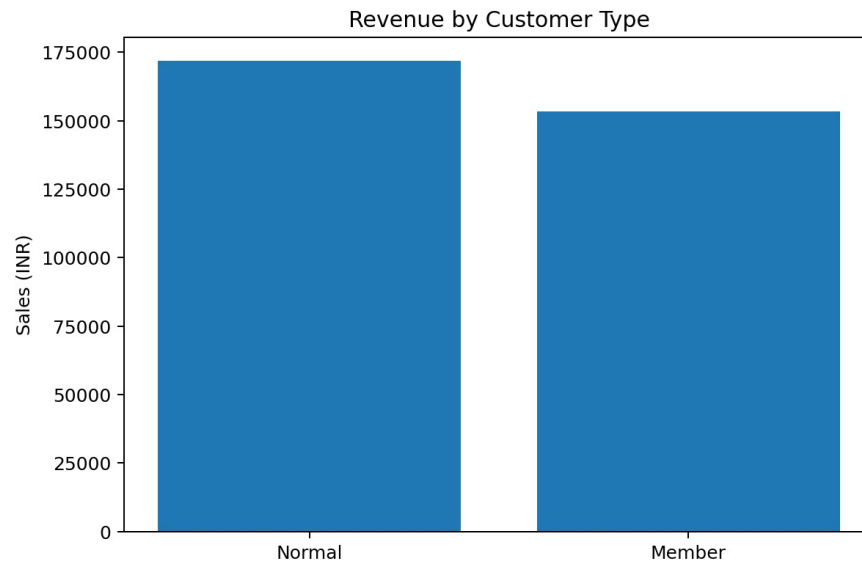


Figure 5. Revenue comparison by customer type.

12.1 PAYMENT METHOD MIX

Ewallet: 438 transactions (36.5%).

Cash: 384 transactions (32.0%).

Credit card: 379 transactions (31.6%).

Payment Method Mix

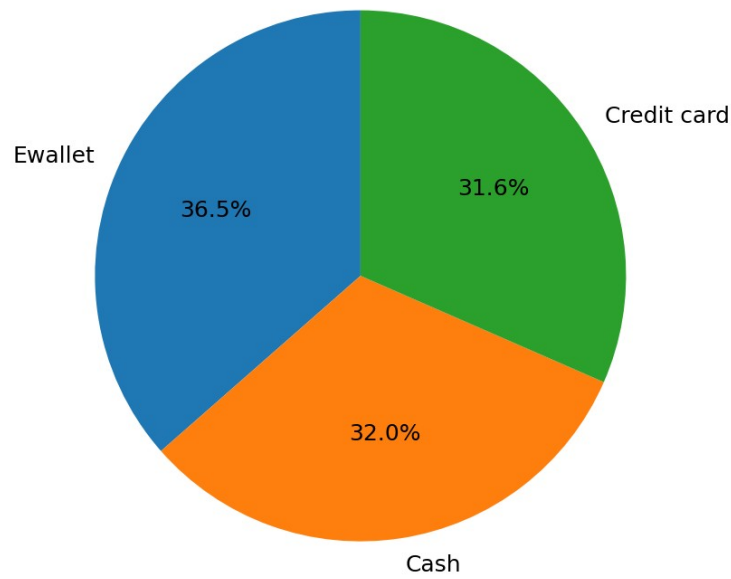


Figure 6. Payment method distribution.

The most frequently used payment method is Ewallet, representing 36.5% of the transactions in the generated dataset.

13. AI-BASED ANOMALY DETECTION

The project uses Isolation Forest to identify statistically unusual transactions. Five standardized features are supplied to the model: Unit_Price, Quantity, Total, Gross_Income, and Rating. The model is configured with contamination = 0.03, so approximately 3% of records are expected to be flagged.

In the cleaned demonstration data, 36 of 1201 transactions are flagged (3.0%). The highest-priority examples include large transaction totals and/or unusually high quantities. The model score provides a ranking for review; it does not establish fraud, billing error, or misconduct.

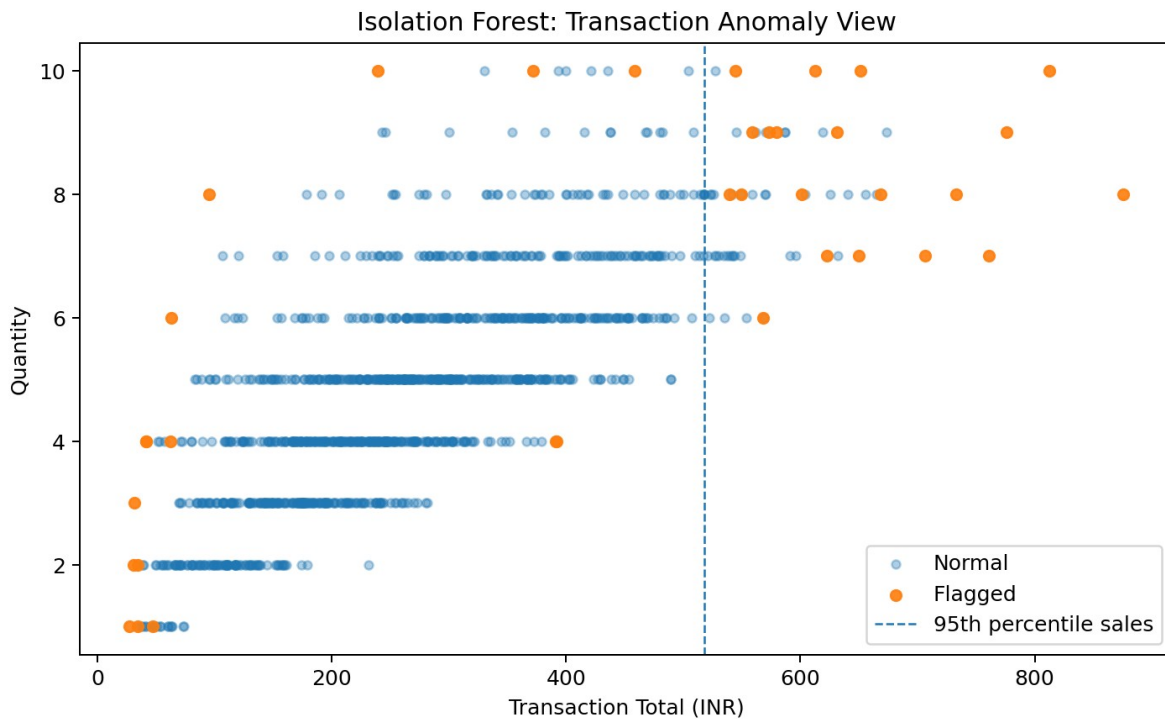


Figure 7. Illustration of flagged transactions under the implemented Isolation Forest model.

Invoice	Date	Branch	Product line	Total	Qty
INV-00015	2026-01-17	B	Electronic accessories	₹458.65	10
INV-00037	2026-04-19	B	Electronic accessories	₹650.08	7
INV-00064	2026-07-20	C	Sports and travel	₹622.66	7
INV-00101	2026-06-16	A	Health and beauty	₹239.53	10
INV-00102	2026-02-09	A	Electronic accessories	₹372.06	10
INV-00132	2026-03-03	A	Electronic accessories	₹63.00	6
INV-00143	2026-06-21	A	Health and beauty	₹579.35	9
INV-00146	2026-04-03	A	Food and beverages	₹34.39	1

14. CUSTOMER SEGMENTATION

The project includes a customer-segmentation function built around aggregated customer-type features: transaction count, revenue, average basket, and average rating. When at least three customer groups are available, the implementation scales these features and runs K-Means with three clusters. In the present demonstration data, only two customer types exist: Member and Normal.

To avoid forcing an unstable three-cluster model on two groups, the current implementation returns stable rule-based labels: “Loyal / higher engagement” for Member and “General shopper” for Normal. This is a deliberate design choice for a small, category-based dataset.

Customer type	Transactions	Revenue	Avg basket	Avg rating	Implemented label
Member	578	₹153,284.55	₹265.20	7.12	Loyal / higher engagement
Normal	623	₹171,722.32	₹275.64	7.10	General shopper

Future customer segmentation should use a true customer identifier, purchase frequency, recency, monetary value, category diversity, and other customer-level features. Without a stable customer ID, transaction rows should not be interpreted as individual customer histories.

15. SYSTEM ARCHITECTURE AND DASHBOARD

The application uses a modular Python structure.

Layer	File / component	Responsibility
Presentation	app.py	Streamlit filters, KPIs, charts, AI insight display, download
Preprocessing	src/preprocessing.py	Validation, type conversion, imputation, feature engineering
AI analytics	src/ai_models.py	Isolation Forest, customer grouping, insight generation
Data	data/supermarket_sales.csv	Demonstration transaction dataset
Documentation	docs/	Report, presentation outline, viva questions

Dashboard capabilities

- Branch, product-line, and customer-type filters
- Sales, transactions, average basket, and rating cards
- Daily sales trend and branch comparison
- Product-line revenue visualization
- AI-generated plain-language insights
- Anomaly review table
- Customer-type summary and payment mix
- CSV download of the filtered cleaned data

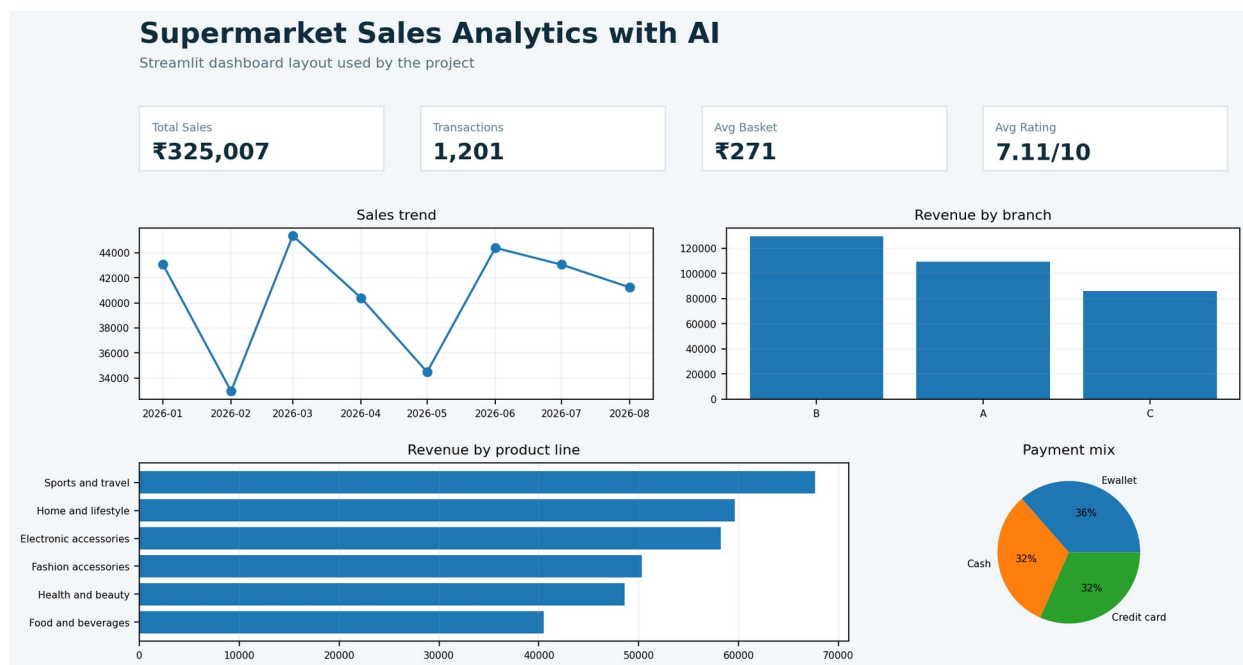


Figure 8. Dashboard composition based on the Streamlit application.

16. RESULTS AND BUSINESS INSIGHTS

The project converts the analytical outputs into concise business-oriented statements. These statements are generated from observed data rather than from unsupported assumptions.

Area	Finding
Branch performance	Branch B (Indore) has the highest total sales: ₹129,669.80.
Product contribution	Sports and travel is the highest-revenue product line: ₹67,657.37.
Payment behavior	Ewallet is the most-used payment method: 438 transactions (36.5%).
Recent trend	The latest 7 represented dates average 7.4% lower daily sales than the preceding 7 represented dates.
AI review queue	36 transactions are flagged by Isolation Forest for manual review.

Example decision-support questions that the dashboard enables

- Which branch contributes the most revenue in the current filtered view?
- Which product lines deserve a closer look for inventory or promotion planning?
- What payment methods dominate transaction volume?
- Which records should a manager inspect first because the model considers them unusual?

These questions support human review. They do not automatically prescribe an action.

17. LIMITATIONS

- The bundled dataset is synthetic/demo data rather than the official internship workbook.
- The current data contains only two customer types and does not provide a persistent customer identifier.
- Isolation Forest is an outlier detector, not a fraud classifier; flagged rows require human investigation.
- The latest-7-day trend is calculated over the represented dates in the generated dataset, not necessarily a continuous real-world calendar.
- The dashboard provides descriptive analytics rather than causal inference.
- No database, authentication, scheduled refresh, model registry, or monitoring layer is included.
- The analytical results depend on the selected filtering context and the quality of the input data.

Interpretation guideline

A strong analytics project reports both what the model found and what the model cannot prove. This report therefore treats anomaly flags as review candidates and branch/product differences as descriptive comparisons.

18. FUTURE SCOPE AND CONCLUSION

Future Scope

- Add weekly/monthly sales forecasting with time-series models.
- Add RFM-based customer analytics when a stable customer ID is available.
- Introduce product recommendation and cross-sell analysis.
- Connect the dashboard to MySQL or PostgreSQL for persistent storage.
- Add authentication and role-based dashboards for managers and analysts.
- Deploy the application to a cloud platform with scheduled data refresh.
- Add model monitoring, drift checks, and an auditable anomaly-review workflow.

Conclusion

The Supermarket Sales Analysis with AI project demonstrates an end-to-end analytics workflow appropriate for an academic and internship setting. It covers data cleaning, feature engineering, exploratory analysis, visualization, anomaly detection, customer summaries, and dashboard communication in one modular application. The project also demonstrates an important practical lesson: the quality of a business insight depends on both the analytical method and the quality of the underlying data.

For final internship submission, the recommended next step is to replace the demonstration CSV with the approved workbook dataset, rerun the pipeline, update the charts and KPIs, and retain the same report structure so that all results remain traceable to the actual source data.

19. REFERENCES AND APPENDIX

References

- AICTE | IBM SkillsBuild Data Analytics with AI Internship Program 2026 - Masterclass Recordings and Resources, supplied PDF attachment (program dates, focus, session schedule, and project resource).
- Python documentation and standard library usage within the project code.
- Pandas documentation for data loading, grouping, aggregation, and missing-value handling.
- Scikit-learn documentation for IsolationForest, KMeans, and StandardScaler.
- Streamlit documentation for interactive data applications.

Appendix A - Project structure

```
supermarket_sales_ai_project/  
├─ app.py  
├─ requirements.txt  
├─ data/supermarket_sales.csv  
├─ data/generate_data.py  
├─ src/preprocessing.py  
├─ src/ai_models.py  
└─ docs/ (report, presentation outline, viva questions)
```

Appendix B - Key implementation command

```
python -m venv .venv  
.venv\Scripts\activate # Windows  
pip install -r requirements.txt  
streamlit run app.py
```

Appendix C - Viva one-line summary

“This project cleans supermarket transaction data, analyzes sales and customer behavior, applies Isolation Forest for unusual transactions, and presents the results in an interactive Streamlit dashboard.”